

Chapter 5

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1 The Derivative of a Real Function

Def 1. Let f be defined (and real-valued) on $[a, b]$. For any $x \in [a, b]$, form the quotient

$$\Phi(t) = \frac{f(t) - f(x)}{t - x} \quad (a < t < b, t \neq x),$$

and define

$$f'(x) = \lim_{t \rightarrow x} \Phi(t),$$

provided this limit exists in accordance with the definition in the last chapter.

We thus associate with the function f a function f' whose domain is the set of points x at which the limit above exists; f' is called the *derivative* of f .

If f' is defined at a point x , we say that f is *differentiable* at x . If f' is defined at every point of a set $E \subset [a, b]$, we say that f is differentiable on E .

Thm 2. Let f be defined on $[a, b]$. If f is differentiable at a point $x \in [a, b]$, then f is continuous at x .

Thm 3. Suppose f and g are defined on $[a, b]$ and are differentiable at a point $x \in [a, b]$. Then $f + g$, fg , and f/g are differentiable at x , and

- (a) $(f + g)'(x) = f'(x) + g'(x)$;
- (b) $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$;
- (c) $(\frac{f}{g})'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$.

In (c), we assume of course that $g(x) \neq 0$.

Thm 4. Suppose f is continuous on $[a, b]$, $f'(x)$ exists at some point $x \in [a, b]$, g is defined on an interval I which contains the range of f , and g is differentiable at the point $f(x)$. If

$$h(t) = g(f(t)) \quad (a \leq t \leq b),$$

then h is differentiable at x , and

$$h'(x) = g'(f(x))f'(x).$$

2 Mean Value Theorems

Def 5. Let f be a real function defined on a metric space X . We say that f has a *local maximum* at a point $p \in X$ if there exists $\delta > 0$ such that $f(q) \leq f(p)$ for all $q \in X$ with $d(p, q) < \delta$. Local minima are defined likewise.

Thm 6. Let f be defined on $[a, b]$; if f has a local maximum at a point $x \in (a, b)$ and if $f'(x)$ exists, then $f'(x) = 0$.

The analogous statement for local minima is of course true.

Thm 7. (*Generalized Mean Value Theorem*) If f and g are continuous real functions on $[a, b]$ which are differentiable on (a, b) , then there is a point $x \in (a, b)$ at which

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

Note that differentiability is not required at end points.

Thm 8. (*The Mean Value Theorem*) If f is a real continuous function on $[a, b]$ which is differentiable on (a, b) , then there is a point $x \in (a, b)$ at which

$$f(b) - f(a) = (b - a)f'(x).$$

Thm 9. Suppose f is differentiable on (a, b) .

- (a) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.
- (b) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.
- (c) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

3 The Continuity of Derivatives

Thm 10. Suppose f is a real differentiable function on $[a, b]$ and suppose $f'(a) < \lambda < f'(b)$. Then there is a point $x \in (a, b)$ such that $f'(x) = \lambda$.

A similar result holds of course if $f'(a) > f'(b)$.

Cor 11. If f is differentiable on $[a, b]$, then f' cannot have any simple discontinuities on $[a, b]$.

But f' may very well have discontinuities of the second kind.

4 L'hospital's Rule

Thm 12. Suppose f and g are real and differentiable on (a, b) , and $g'(x) \neq 0$ for all $x \in (a, b)$, where $-\infty \leq a < b \leq +\infty$. Suppose

$$\frac{f'(x)}{g'(x)} \rightarrow A \text{ as } x \rightarrow a.$$

If

$$f(x) \rightarrow 0 \text{ and } g(x) \rightarrow 0 \text{ as } x \rightarrow a,$$

or if

$$g(x) \rightarrow +\infty \text{ as } x \rightarrow a,$$

then

$$\frac{f(x)}{g(x)} \rightarrow A \text{ as } x \rightarrow A.$$

5 Derivatives of Higher Order

Def 13. If f has a derivative f' on an interval, and if f' is itself differentiable, we denote the derivative of f' by f'' and call f'' the second derivative of f . Continuing in this manner, we obtain functions

$$f, f', f'', f^{(3)}, \dots, f^{(n)},$$

each of which is the derivative of the preceding one. $f^{(n)}$ is called the n th derivative of f , or the derivative of order n of f .

6 Taylor's Theorem

Thm 14. Suppose f is a real function on $[a, b]$, n is a positive integer, $f^{(n-1)}$ is continuous on $[a, b]$, $f^{(n)}(t)$ exists for every $t \in (a, b)$. Let α, β be distinct points of $[a, b]$, and define

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k.$$

Then there exists a point x between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n.$$

7 Differentiation of Vector-Valued Functions

Thm 15. Suppose $\mathbf{f}$ is a continuous mapping of $[a, b]$ into $\mathbb{R}^k$ and $\mathbf{f}$ is differentiable on (a, b) . Then there exists $x \in (a, b)$ such that

$$|\mathbf{f}(b) - \mathbf{f}(a)| \leq (b - a)|\mathbf{f}'(x)|.$$

8 Exercise Notes

Thm 16. Suppose g is the inverse function of f and is differentiable, then

$$g'(f(x)) = \frac{1}{f'(x)}$$

Thm 17. Suppose $f'(x), g'(x)$ exist, $g'(x) \neq 0$, and $f(x) = g(x) = 0$. Then

$$\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}.$$

(This also holds for complex functions.)

Thm 18. Suppose f' is continuous on $[a, b]$ and $\varepsilon > 0$. Prove that there exists $\delta > 0$ such that

$$\left| \frac{f(t) - f(x)}{t - x} - f'(x) \right| < \varepsilon$$

whenever $0 < |t - x| < \delta$, $a \leq x \leq b$, $a \leq t \leq b$. This holds for vector-valued functions too.

Thm 19. Suppose $a \in \mathbb{R}^1$, f is a twice-differentiable real function on (a, ∞) , and M_0, M_1, M_2 are the least upper bounds of $|f(x)|, |f'(x)|, |f''(x)|$, respectively, on (a, ∞) . Prove that

$$M_1^2 \leq 4M_0M_2.$$

Rmk 20. The theorem above is true for vector-valued functions.

Thm 21. Suppose f is defined in $(-1, 1)$ and $f'(0)$ exists. Suppose $-1 < \alpha_n <$

$\beta_n < 1$, $\alpha_n \rightarrow 0$, and $\beta_n \rightarrow 0$ as $n \rightarrow \infty$. Define the difference quotients

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n}.$$

Then

- (a) If $\alpha_n < 0 < \beta_n$, then $\lim D_n = f'(0)$.
- (b) If $0 < \alpha_n < \beta_n$ and $(\beta_n/(\beta_n - \alpha_n))_{n=1}^{\infty}$ is bounded, then $\lim D_n = f'(0)$.
- (c) If f' is continuous in $(-1, 1)$, then $\lim D_n = f'(0)$.

Thm 22. For vector-valued function $f : [a, b] \rightarrow \mathbb{R}^m$ which $f^{(n)}$ exists on (a, b) . For $\alpha, \beta \in [a, b]$, define

$$M_n = \sup_{\xi \text{ between } \alpha, \beta} \|f^{(n)}(\xi)\|.$$

Then

$$\|f(\beta) - \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (\beta - \alpha)^k\| \leq \frac{M_n}{n!} |\beta - \alpha|^n.$$